

C992

College Geometry

Armored Drill

8 concepts · 12 items · 4 transfer items

What this is. Every concept in this course's public description, re-asked with more disguise than a practice test uses. Each stem carries facts that do not decide it. Every wrong answer is a named trap.

Source. Scope from the course's public description in the WGU Institutional Catalog (September 2026); items written from the field's standard body of knowledge. No WGU assessment item, course material or proprietary content is used or reproduced. Not affiliated with or endorsed by Western Governors University.

Method © 2026 Erik Daniel Lindquist — Course Mastery Coach, armored drill. Items drafted with AI assistance under his direction.

How to run it

Paper first. Pencil. One pass, no notes, a timer if you want the pressure.

1. **Before you read the options**, fill the strip under the item: the target word (what is actually being asked), and the one fact in the stem that decides it.
2. **Principle before arithmetic**. Name the rule, then compute.
3. **Camouflage**. Most of every stem does not matter. Ask: which single fact decides this?
4. **Inverse pairs are the bait**. If two options are opposites, one of them is the trap built for you.
5. **Mark sure, unsure or guess** honestly. Sure-and-wrong is the most valuable miss on the page — it is a wrong mental model.
6. **Absolutes** (always, never, guarantee) are usually wrong. And recognizing an answer is not the same as owning it.
7. Items with a **heavy black bar** on the left are **transfer items**: the same principle, reversed or solved for a different variable.
8. Stop at the STOP page. Then grade, fill the triage grid, and tally the traps you fell for. The biggest tally is what you drill next.

The strip, filled in, looks like this:

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

Items

1.

A geometry student working with dynamic software on a shared lab computer studies a four-axiom system. She builds a model with five points and some lines where axioms 1, 2 and 3 hold but axiom 4 fails. Her instructor has written two textbooks on axiomatic systems. What does this model prove?

- A. Axiom 4 is false in every model.
- B. Axiom 4 is independent of axioms 1, 2 and 3.
- C. The four-axiom system is consistent.
- D. The four-axiom system is inconsistent.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

2.

A textbook written by a well-known author states: “If a quadrilateral is a square, then it is a rectangle.” A class discussing the statement notes that squares appear often in architecture. Which statement is logically equivalent?

- A. If a quadrilateral is not a square, then it is not a rectangle.
- B. If a quadrilateral is a rectangle, then it is a square.
- C. If a quadrilateral is not a rectangle, then it is not a square.
- D. Every rectangle is a square.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

3. [TRANSFER — reversed stem / different variable]

The statement “If a triangle is equilateral, then it is isosceles” is true. A student claims its converse is also true. Which is a valid counterexample to the converse?

- A. A triangle with sides 6, 6 and 6.
- B. A triangle with sides 3, 4 and 5.
- C. No counterexample exists because the original is true.
- D. A triangle with sides 5, 5 and 8.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

4.

Two triangular roof trusses built by the same contractor from the same lumber each have sides of 8 ft and 10 ft and a 40° angle opposite the 8-ft side. Which conclusion is justified?

- A. They are congruent because they came from the same lumber and builder.
- B. The trusses are not necessarily congruent; SSA does not prove congruence.
- C. They are similar but never congruent.
- D. They are congruent by SAS.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

5.

A surveyor using new GPS equipment plots a lot with corners at (0, 0), (4, 0), (5, 3) and (1, 3). The lot sits beside a park. What is the most specific name for the quadrilateral?

- A. Parallelogram.
- B. Rhombus.
- C. Trapezoid with only one pair of parallel sides.
- D. Rectangle.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

6.

In a design app with a sleek new interface, a logo is reflected across the line $x = 1$ and then across the line $x = 4$. The logo is blue and 2 units wide. What single transformation gives the same result?

- A. A translation 6 units to the left.
- B. A translation 6 units to the right.
- C. A rotation of 180° .
- D. A translation 3 units to the right.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

7. [TRANSFER — reversed stem / different variable]

A figure is reflected across two lines that intersect at point P and form a 35° angle. What single transformation is equivalent?

- A. A rotation of 70° about P.
- B. A rotation of 17.5° about P.
- C. A rotation of 35° about P.
- D. A translation of 70 units.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

8.

On a globe in a campus lab, a student draws a triangle from the North Pole down to the equator, 90° of longitude along the equator, and back up. Each angle measures 90° . The globe is 30 cm across. What does the angle sum show?

- A. The sum is 180° , since every triangle's angles sum to 180° .
- B. The sum is 270° , consistent with hyperbolic geometry.
- C. The sum is 270° , consistent with spherical (elliptic) geometry, where sums exceed 180° .
- D. The sum depends on the globe's 30 cm size.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

9. [TRANSFER — reversed stem / different variable]

In hyperbolic geometry, through a point not on a given line, how many lines are parallel to the given line?

- A. None.
- B. Exactly one.
- C. Infinitely many.
- D. Exactly one, because parallels are unique in every geometry.

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

10.

Three villages, each with a growing population and a weekly market, want to drill one shared well the same distance from all three village centers. Which construction locates the well?

- A. The intersection of the angle bisectors (the incenter).
- B. Next to the village with the largest market.
- C. The intersection of the medians (the centroid).
- D. The intersection of the perpendicular bisectors of the triangle's sides (the circumcenter).

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

11.

A designer building a light fixture for a hotel lobby uses a convex polyhedron with 12 vertices and 30 edges. Each face will be frosted glass. How many faces does it have?

- A. 20
- B. 18
- C. 16
- D. 44

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	___	

12. [TRANSFER — reversed stem / different variable]

A convex polyhedron has 8 faces and 6 vertices. How many edges does it have?

- A.** 16
- B.** 8
- C.** 14
- D.** 12

Target word	The one fact that decides	Answer	[] sure [] unsure [] guess
_____	_____	_____	

STOP

Answers and dissection follow. Finish every item and every strip first.

Key and dissection

Struck-through text is the camouflage: facts that do not decide the item.

1. Answer B Concept 1

A geometry student ~~working with dynamic software on a shared lab computer~~ studies a four-axiom system. She builds a model with five points and some lines where axioms 1, 2 and 3 hold but axiom 4 fails. ~~Her instructor has written two textbooks on axiomatic systems.~~ What does this model prove?

Target word: what the model proves **Decides:** Axioms 1–3 true and axiom 4 false in one model.

A	Absolute language	One model where it fails does not mean it fails everywhere.
B	Correct	If axiom 4 could be proved from the others, it would have to hold in every model where they hold.
C	Adjacent category	Consistency needs a model where all four axioms hold.
D	Inverted operation	Inconsistency means the axioms contradict each other; a model where one fails says nothing about that.

2. Answer C Concept 2

A textbook ~~written by a well-known author~~ states: “If a quadrilateral is a square, then it is a rectangle.” ~~A class discussing the statement notes that squares appear often in architecture.~~ Which statement is logically equivalent?

Target word: equivalent statement **Decides:** Only the contrapositive is equivalent.

A	Confusion set	The inverse; also false for a non-square rectangle.
B	Inverted operation	The converse; false for a non-square rectangle.
C	Correct	The contrapositive always has the same truth value.
D	Absolute language	Restates the false converse as a universal claim.

3. Answer D Concept 2 · transfer

The statement “If a triangle is equilateral, then it is isosceles” is true. A student claims its converse is also true. Which is a valid counterexample to the converse?

Target word: counterexample to the converse **Decides:** Converse: if isosceles, then equilateral. Need isosceles but not equilateral.

A	Inverted operation	Satisfies both parts; it supports, not refutes, the converse.
B	Dropped element	Not isosceles, so it does not meet the converse’s hypothesis.
C	Absolute language	A true statement can have a false converse.
D	Correct	Isosceles but not equilateral, so the converse fails.

4. Answer B Concept 3

Two triangular roof trusses ~~built by the same contractor from the same lumber~~ each have sides of 8 ft and 10 ft and a 40° angle opposite the 8-ft side. Which conclusion is justified?

Target word: is congruence proved **Decides:** The angle is not between the given sides: SSA.

A	Camouflage	Materials and builder do not decide congruence.
B	Correct	SSA allows two different triangles (the ambiguous case).
C	Absolute language	They might be congruent; the data just do not prove it.
D	Confusion set	SAS needs the angle between the two sides.

5. Answer A Concept 4

A surveyor ~~using new GPS equipment~~ plots a lot with corners at (0, 0), (4, 0), (5, 3) and (1, 3). ~~The lot sits beside a park.~~ What is the most specific name for the quadrilateral?

Target word: most specific classification **Decides:** Opposite sides parallel (slopes 0 and 3), adjacent sides not perpendicular, side lengths 4 and $\sqrt{10}$.

A	Correct	Both pairs of opposite sides are parallel; slopes 0 and 3 are not perpendicular, and sides 4 and $\sqrt{10}$ are unequal.
B	Confusion set	Needs four equal sides; $4 \neq \sqrt{10}$.
C	Adjacent category	Both pairs of opposite sides are parallel.
D	Dropped element	Skips the perpendicularity check; slopes 0 and 3 do not meet at a right angle.

6. Answer B Concept 5

In a design app ~~with a sleek new interface~~, a logo is reflected across the line $x = 1$ and then across the line $x = 4$. ~~The logo is blue and 2 units wide.~~ What single transformation gives the same result?

Target word: equivalent single transformation **Decides:** Parallel lines 3 units apart $\rightarrow$ translation of twice that distance.

A	Inverted operation	Right amount, wrong direction.
B	Correct	$2 \times 3 = 6$, in the direction from the first line to the second; e.g., $(0,0) \rightarrow (2,0) \rightarrow (6,0)$.
C	Adjacent category	Rotations come from intersecting lines, not parallel ones.
D	Dropped element	Forgets to double the distance.

7. Answer A Concept 5 • transfer

A figure is reflected across two lines that intersect at point P and form a 35° angle. What single transformation is equivalent?

Target word: equivalent single transformation **Decides:** Intersecting lines $\rightarrow$ rotation of twice the angle.

A	Correct	$2 \times 35^\circ = 70^\circ$.
B	Inverted operation	Halves instead of doubles.
C	Dropped element	Forgets to double the angle.
D	Adjacent category	Translations come from parallel lines.

8. Answer C Concept 6

On a globe ~~in a campus lab~~, a student draws a triangle from the North Pole down to the equator, 90° of longitude along the equator, and back up. Each angle measures 90° . ~~The globe is 30 cm across.~~ What does the angle sum show?

Target word: which geometry fits **Decides:** Angle sum is 270° , more than 180° .

A	Absolute language	True only in Euclidean geometry.
B	Confusion set	Hyperbolic sums are less than 180° .
C	Correct	On a sphere, triangle angle sums are greater than 180° .
D	Camouflage	For this triangle each angle is 90° on any size sphere.

9. Answer C Concept 6 • transfer

In hyperbolic geometry, through a point not on a given line, how many lines are parallel to the given line?

Target word: number of parallels **Decides:** Hyperbolic replaces the parallel postulate with more than one parallel.

A	Confusion set	That is spherical (elliptic) geometry.
B	Confusion set	That is Euclidean geometry's parallel postulate.
C	Correct	Hyperbolic geometry allows infinitely many parallels through the point.
D	Absolute language	Uniqueness depends on which parallel postulate is chosen.

10. Answer D Concept 7

Three villages, ~~each with a growing population and a weekly market~~, want to drill one shared well the same distance from all three village centers. Which construction locates the well?

Target word: which construction **Decides:** Equidistant from three points → circumcenter.

A	Confusion set	Equidistant from the sides, not the vertices.
B	Camouflage	Market size does not decide equal distance.
C	Adjacent category	The balance point; not equidistant from the vertices in general.
D	Correct	Every point on a perpendicular bisector is equidistant from that side's endpoints.

11. Answer A Concept 8

A designer building a light fixture ~~for a hotel lobby~~ uses a convex polyhedron with 12 vertices and 30 edges. ~~Each face will be frosted glass.~~ How many faces does it have?

Target word: number of faces **Decides:** $V - E + F = 2$.

A	Correct	$F = 2 - 12 + 30 = 20$ (an icosahedron).
B	Dropped element	$30 - 12$ leaves out the 2.
C	Inverted operation	$30 - 12 - 2$ subtracts the 2 instead of adding it.
D	Inverted operation	$12 + 30 + 2$ adds every term.

12. Answer D Concept 8 • transfer

A convex polyhedron has 8 faces and 6 vertices. How many edges does it have?

Target word: solve for edges **Decides:** $E = V + F - 2$.

A	Inverted operation	$6 + 8 + 2$ adds the 2.
B	Wrong stem number	The number of faces.
C	Dropped element	$6 + 8$ leaves out the $- 2$.
D	Correct	$6 + 8 - 2 = 12$ (an octahedron).

Triage grid

Write each item number in the box it belongs to.

	Right	Wrong
Sure	Owled.	Concept inversion — top priority. A wrong mental model; it survives rewording.
Unsure / guess	Recognition, not mastery. Re-drill with the transfer.	Not learned yet.

Trap tally

For every miss, find the letter you chose in the key and tally its trap. The biggest number is the pattern to drill.

Trap	What it does	In this drill	Your misses
Dropped element	Leaves out a required term or step.	6	
Wrong stem number	A figure copied from the prompt that is not the answer.	1	
Adjacent category	The right idea in the wrong bucket.	5	
Inverted operation	Direction flipped: add/subtract, cause/effect, higher/lower.	8	
Absolute language	Always, never, guarantee — where judgment applies.	6	
Confusion set	Every option is real; the test is exact role-matching.	7	
Camouflage	Answers a stem fact that was never the question.	3	

Anchor rows — the concept list

One row per concept: the rule to lock in, the trap that catches people, and the tell in the question that signals it.

#	Concept	Rule	Trap	Tell
1	Consistency and independence of axioms	A model where all axioms hold shows the system is consistent; a model where one axiom fails while the others hold shows that axiom is independent of the others.	Thinking a model where an axiom fails proves the system inconsistent.	"A model is found in which..." and a question about what it proves.
2	Contrapositive vs. converse	A conditional is logically equivalent to its contrapositive (if not Q, then not P); its converse (if Q, then P) may be false.	Treating the converse as automatically true.	A true if-then statement and a choice of rewordings.
3	Triangle congruence criteria	SSS, SAS, ASA, AAS and HL (right triangles) prove congruence; SSA and AAA do not.	Accepting two sides and a non-included angle as proof.	The angle given is not between the two sides given.
4	Analytic (coordinate) proof	Classify a figure on the coordinate plane with slopes (parallel: equal; perpendicular: negative reciprocals) and distances; every property must be checked, not assumed from a picture.	Calling a parallelogram a rectangle because it looks right, without checking perpendicularity.	Four vertices given and a question about the most specific name.
5	Composition of reflections	Reflecting across two parallel lines equals a translation of twice the distance between them; across two intersecting lines, a rotation of twice the angle between them about the intersection.	Forgetting to double the distance or angle.	Two reflections in a row.
6	Non-Euclidean geometries	Replacing the parallel postulate changes theorems: in spherical (elliptic) geometry there are no parallels and triangle angle sums exceed 180° ; in hyperbolic geometry there are infinitely many parallels through a point and sums are less than 180° .	Assuming 180° angle sums and unique parallels hold in every geometry.	A triangle drawn on a sphere or a saddle surface.
7	Triangle centers from constructions	Perpendicular bisectors of the sides meet at the circumcenter, equidistant from the vertices; angle bisectors meet at the incenter, equidistant from the sides.	Mixing up which construction gives equal distance to points vs. to sides.	"Equidistant from three points" vs. "equidistant from three roads (lines)."
8	Euler's formula for polyhedra	For any convex polyhedron, $V - E + F = 2$.	Adding all three counts or dropping the 2.	Two of vertices, edges and faces given; the third asked.

Course scope (public description): College Geometry covers the knowledge and skills necessary to use dynamic technology to explore geometry, to use axiomatic reasoning to prove statements about geometry, and to

apply geometric models to solve real-life problems. Topics include axiomatic systems, analytic proofs, coordinate geometry, plane and solid Euclidean geometry, non-Euclidean geometries, constructions, transformations, deductive reasoning, and dynamic technology. College Algebra as well as Trigonometry and Precalculus are prerequisites.